

Homework 8

✧ Signal operation: 7.3/7.5/7.6/7.9/7.10/7.11/7.12/7.17/7.18

7.3 设计一个比例运算电路，要求输入电阻 $R_i = 20\text{k}\Omega$ ，比例系数为-100。

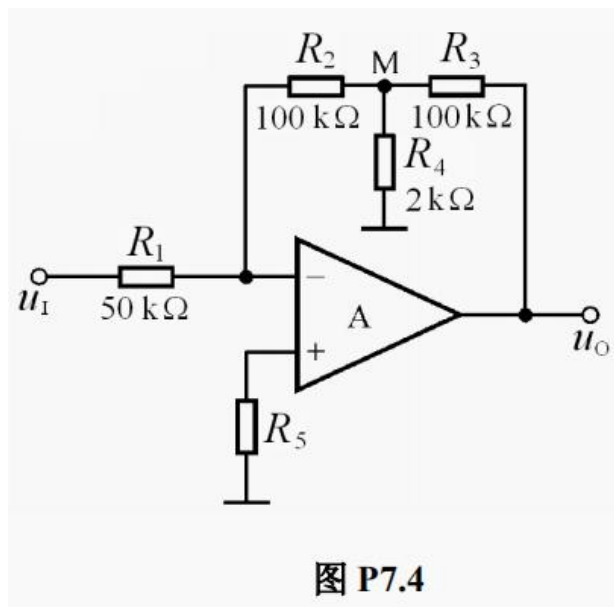
7.3 Design a scaling operation circuit, which has an input resistance $R_i = 20\text{k}\Omega$, and the scaling factor is -100.

7.5 电路如图 P7.4 所示，集成运放输出电压的最大幅值为 $\pm 14\text{V}$ ， u_i 为 2V 的直流信号。分别求出下列各种情况下的输出电压：

(1) R_2 短路； (2) R_3 短路； (3) R_4 短路； (4) R_4 断路。

7.5 As shown in figure p7.4, the maximum amplitude of the output voltage of the integrated operational amplifier is $\pm 14\text{V}$, and the u_i is a 2V DC signal. Calculate the output voltage under the following conditions:

- (1) if $R_2=0$, $u_o=?$
- (2) if $R_3=0$, $u_o=?$
- (3) if $R_4=0$, $u_o=?$
- (4) if $R_4=\infty$, $u_o=?$



7.6 试求图 P7.6 所示各电路输出电压与输入电压的运算关系式。

7.6 Please find the relationship between output voltage and input voltage ($u_o \sim u_i$) of each circuit as shown in figure p7.6.

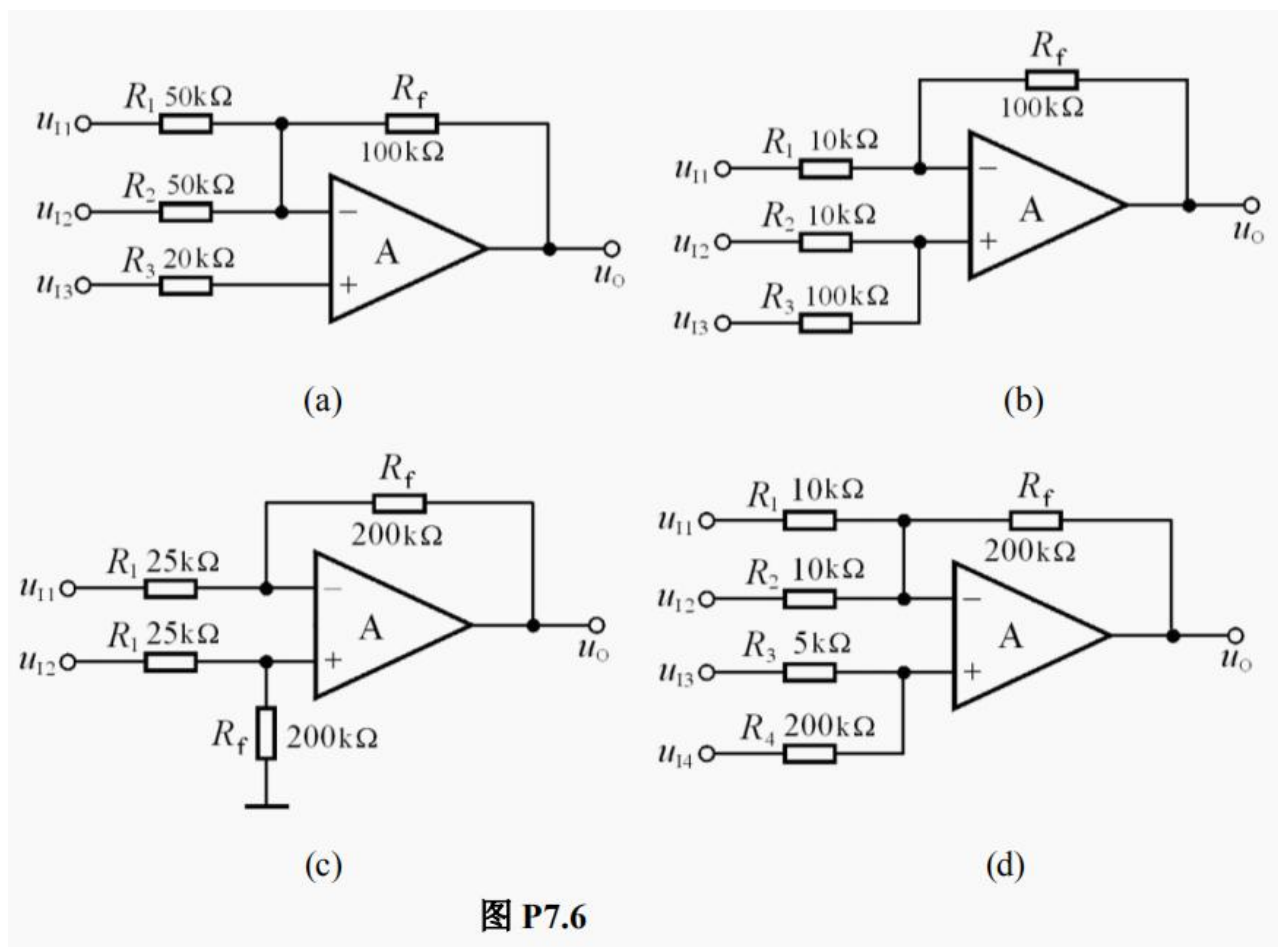


图 P7.6

7.9 电路如图 P7.9 所示。

(1) 写出 U_o 与 U_{i1} 、 U_{i2} 的运算关系式；

(2) 当 R_w 的滑动端在最上端时，若 $U_{i1}=10\text{mV}$ ， $U_{i2}=20\text{mV}$ ，则 $U_o=?$

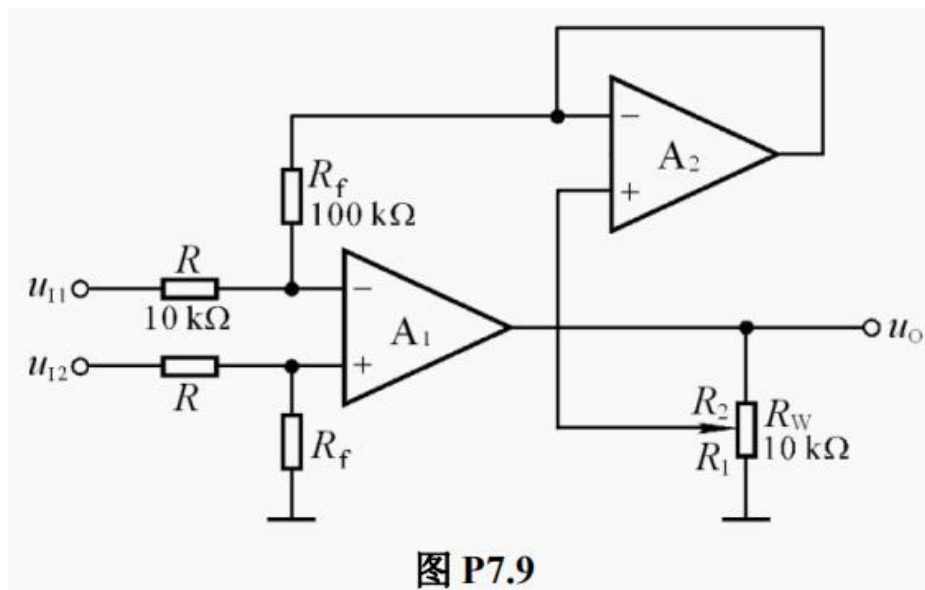
(3) 若 U_o 的最大幅值为 $\pm 14\text{V}$ ，输入电压最大值 $U_{i1\text{max}}=10\text{mV}$ ， $U_{i2\text{max}}=20\text{mV}$ ，它们的最小值均为 0，则为了保证集成运放工作在线性区， R_2 的最大值为多少？

7.9 The circuit is shown in figure p7.9.

(1) Write the operation relation between u_o , u_{i1} and u_{i2} ;

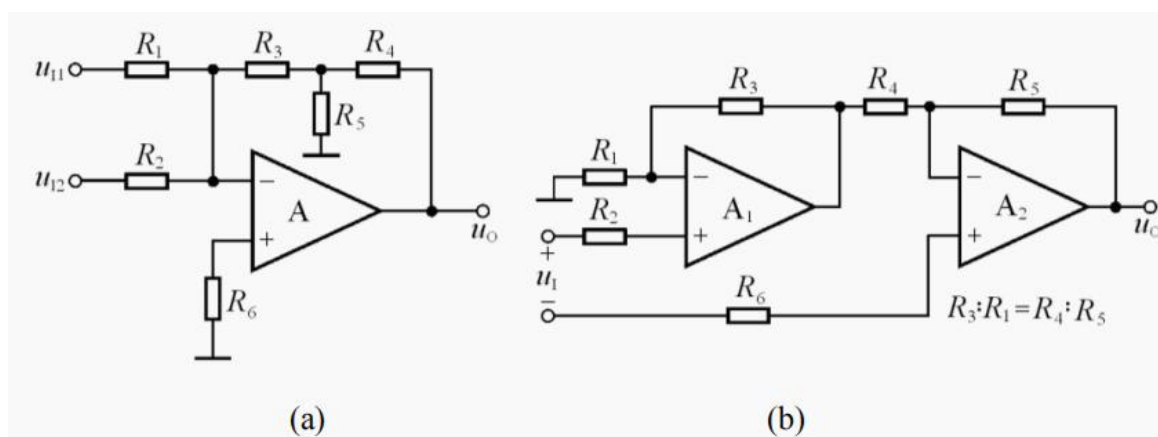
(2) When the sliding end of R_w is at the top, $u_{i1} = 10\text{mV}$, $u_{i2} = 20\text{mV}$, and then $u_o = ?$

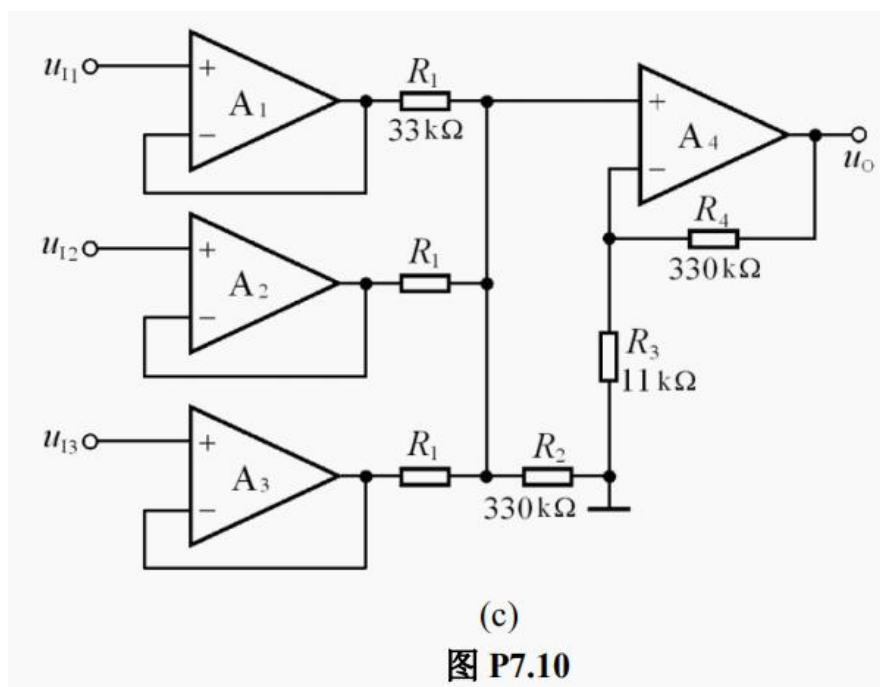
(3) If the maximum amplitude of U_o is $\pm 14\text{V}$, the maximum input voltage is $u_{i1\text{max}} = 10\text{mV}$, $u_{i2\text{max}} = 20\text{mV}$, and their minimum values are all 0, what is the maximum value of R_2 to ensure that the integrated operational amplifier is working in the linear amplification region?



7.10 分别求解图 P7.10 所示各电路的运算关系。

7.10 Solve the operation relationship of each circuit (output voltages as function of input voltages) as shown in figure p7.10.

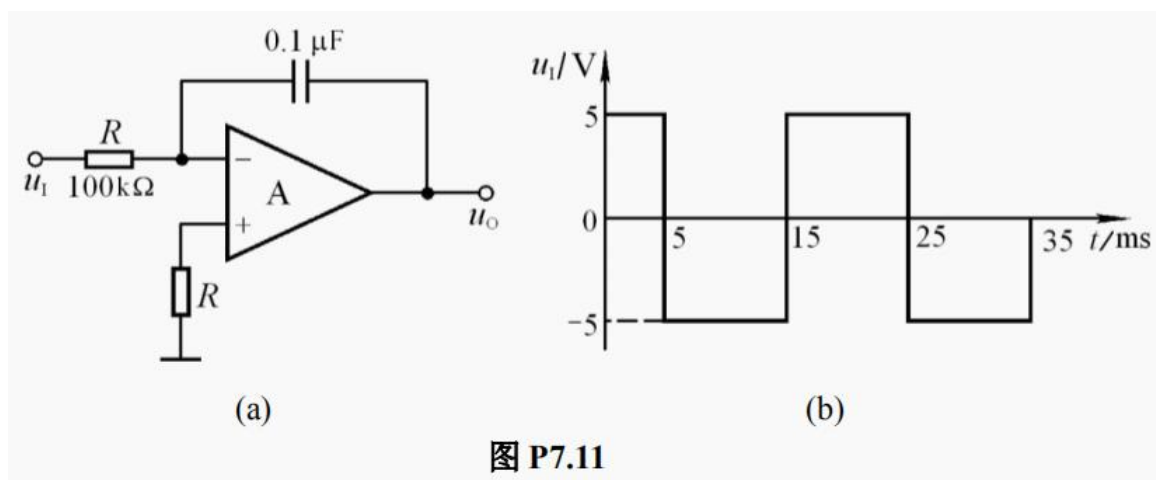




7.11 在图 P7.11(a)所示电路中, 已知输入电压 u_i 的波形如图(b)所示, 当 $t=0$ 时 $u_c=0$ 。试画出输出电压 u_o 的波形。

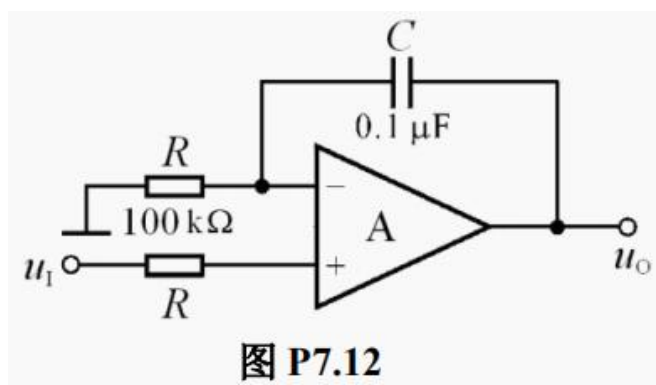
7.11 An operational circuit is shown in figure p7.11 (a), and the waveform of the input voltage u_i is shown in figure p7.11 (b). At $t = 0$, voltage on capacitor $u_c = 0$.

Please draw the waveform of the output voltage u_o .



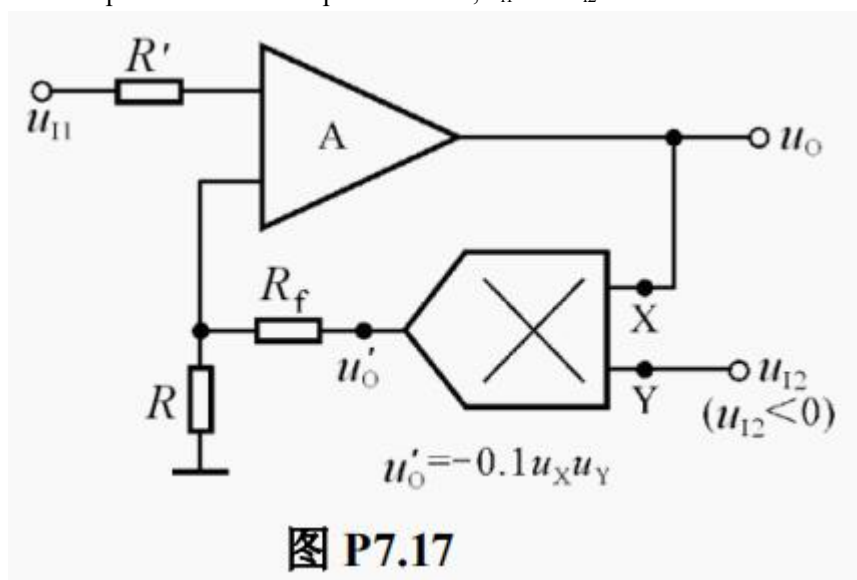
7.12 已知图 P7.12 所示电路输入电压 u_i 的波形如图 P7.11 (b)所示, 且当 $t=0$ 时 $u_c=0$ 。试画出输出电压 u_o 的波形。

7.12 An operational circuit is shown in figure p7.12 (a), and the waveform of the input voltage u_i is shown in figure p7.11 (b). At $t = 0$, voltage on capacitor $u_c = 0$. Please draw the waveform of the output voltage u_o .



7.17 为了使图 P7.17 所示电路实现除法运算，(1)标出集成运放的同相输入端和反相输入端；(2)求出 u_o 和 u_{i1} 、 u_{i2} 的运算关系式。

7.17 In order to realize division operation in the circuit shown in figure p7.17, (1) Mark the non-inverted input terminal and the inverted input terminal of the operational amplifier; (2) Find out the operation relationship between u_o , u_{i1} and u_{i2} .



7.18 求出图 P7.18 所示各电路的运算关系。

7.18 Find the operation relationship (output voltages as function of input voltages) of each circuit as shown in figure p7.18.

